# Detailed versus global legalize

Global legalize lite maps to Tetris shelf pack, displacement six, HPWL thirty-two

# The idea

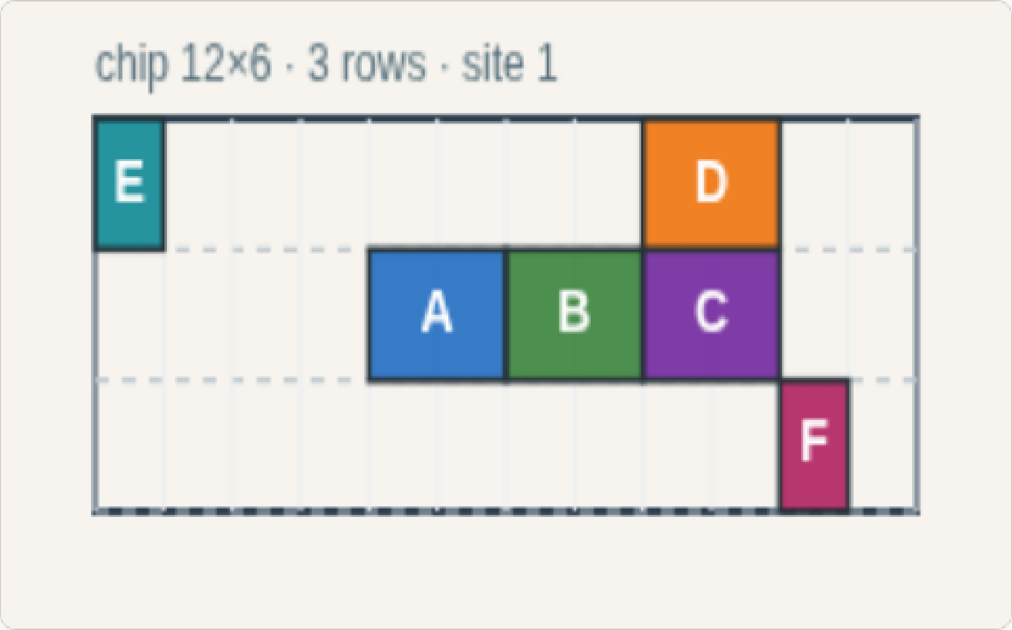
- Pick global Tetris when you want a fast pass and can afford extra movement
- Pick detailed Abacus when displacement budget is tight
- Report both pipelines side by side in regressions, legal first, then disp and HPWL
- <!-- algorithm-walkthrough -->

# Global = Tetris, disp 6

DETAILED VS GLOBAL LEGALIZE

**Global = Tetris, disp 6**

Step 1 / 5 · global-tetris · module04-01-detailed-vs-global




### Explanation

Global legalize lite maps to Tetris-style nearest-row shelf pack. On the overlap seed it legalizes with displacement six and HPWL thirty-two.

- globalLegalize → tetris
- single-row shelf for A,B,C
- fast, simple pipeline stage

```
globalDisp: 6
HPWL: 32
legal: true
```

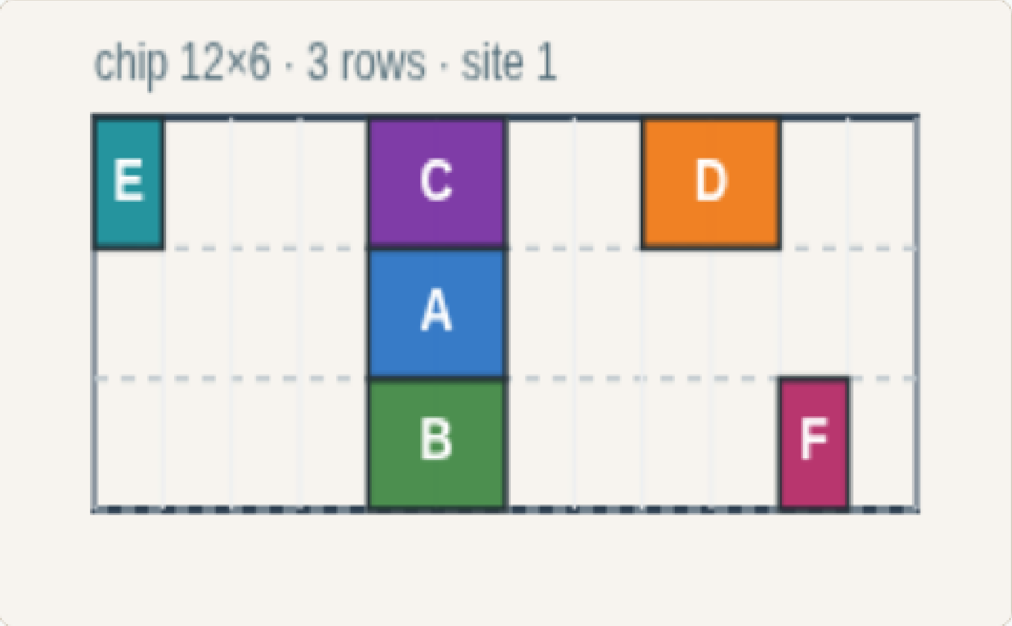
Global = Tetris, disp 6

# Detailed = Abacus, disp 4

DETAILED VS GLOBAL LEGALIZE

## Detailed = Abacus, disp 4

Step 2 / 5 · detailed-abacus · module04-01-detailed-vs-global




### Explanation

Detailed legalize lite maps to Abacus row trial. Same seed, both legal—but displacement drops to four by spreading A, B, C across rows.

- detailedLegalize → abacus
- cross-row assignment
- lower L1 movement

```
detailedDisp: 4
HPWL: 38
legal: true
```

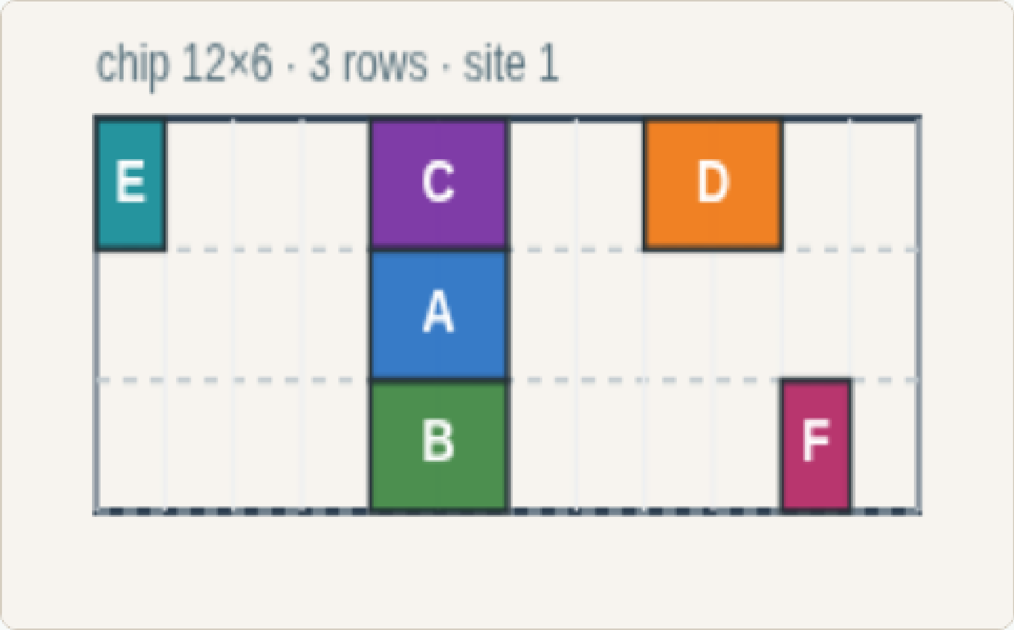
Detailed = Abacus, disp 4

# Both pipelines legal

DETAILED VS GLOBAL LEGALIZE

## Both pipelines legal

Step 3 / 5 · both-legal · module04-01-detailed-vs-global




### Explanation

Global and detailed both pass legality on the overlap seed. The difference is how far cells move and how HPWL shifts—thirty-two versus thirty-eight here.

- Same starter coordinates
- Both isLegal true
- Metrics tell the story

```
global legal: true
detailed legal: true
HPWL delta: 6
```

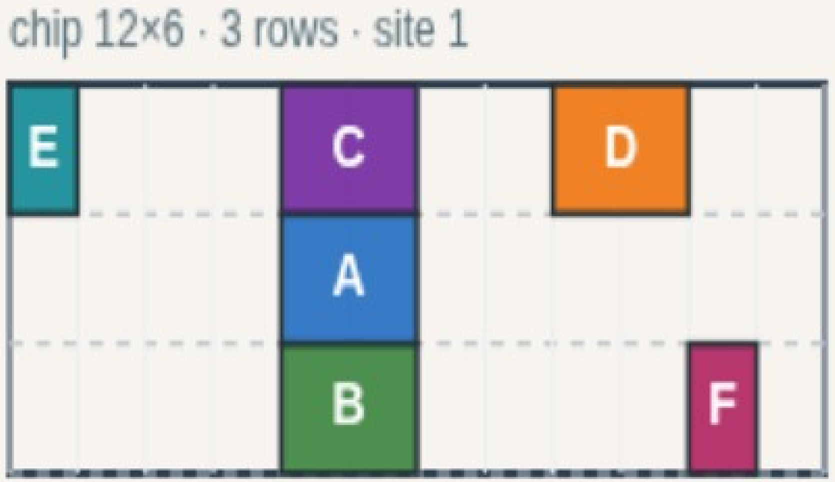
Both pipelines legal

# Side-by-side metrics

DETAILED VS GLOBAL LEGALIZE

## Side-by-side metrics

Step 4 / 5 · side-by-side · module04-01-detailed-vs-global




### Explanation

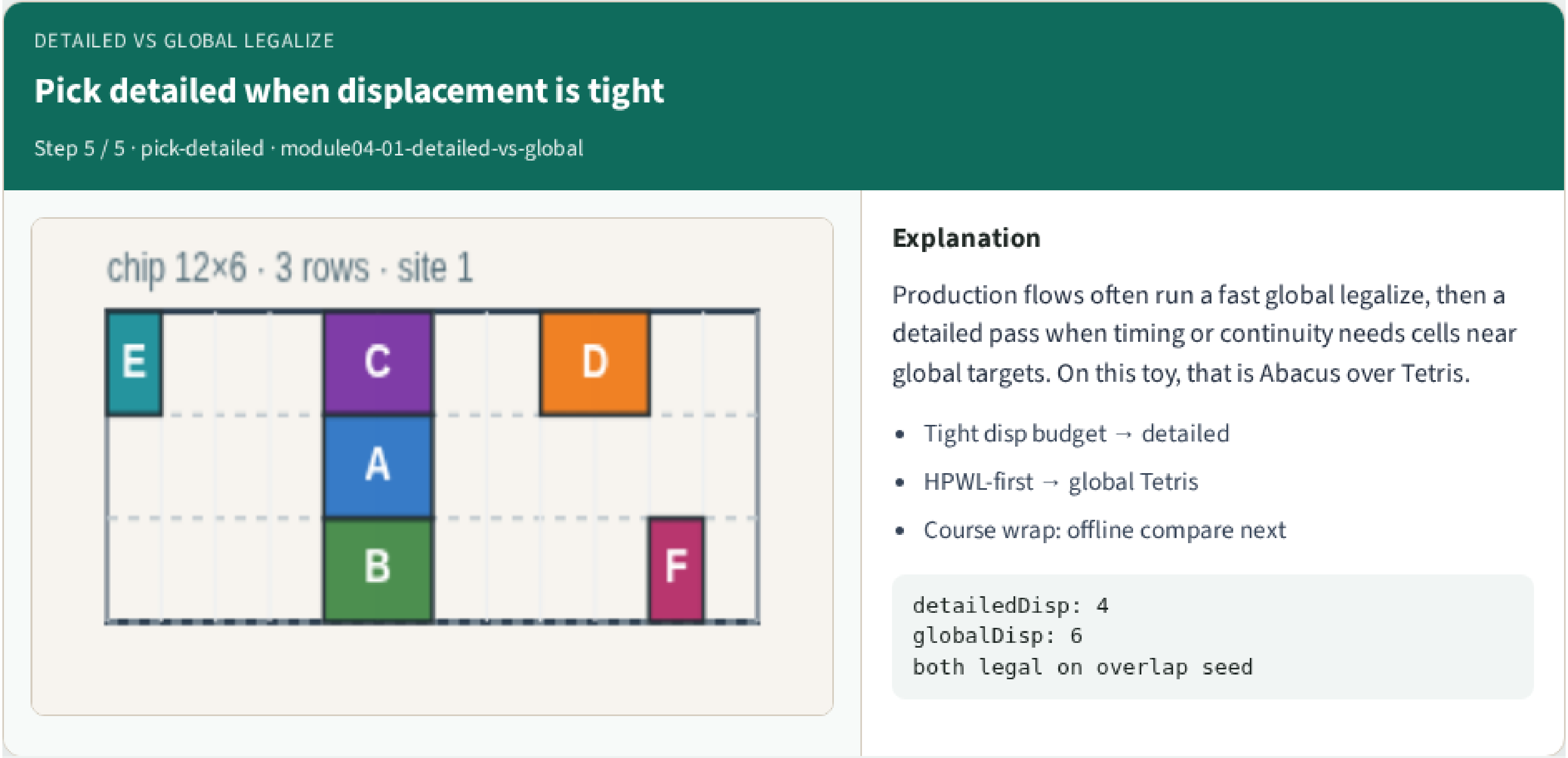
Global Tetris: disp six, HPWL thirty-two. Detailed Abacus: disp four, HPWL thirty-eight. Neither dominates on both axes—pick by your displacement budget.

- global: disp 6 HPWL 32
- detailed: disp 4 HPWL 38
- Report both in regressions

```
globalDisp: 6
detailedDisp: 4
tetrisHpwl: 32
abacusHpwl: 38
```

Side-by-side metrics

# Pick detailed when displacement is tight




Pick detailed when displacement is tight

# Browser lab track

- In the browser lab track, open the **detailed-vs-global** lab from the tools shelf
- Load the overlap or float starter, run the legalizer once
- Work the challenges that lock the goldens

# Implement track

- In the implement track, open this module's examples and the course `common/` solvers
- Parse `tiny_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

# Pitfalls

- Common traps

# Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module